

# Euclidean persistence and sharp stability for anisotropic eigenvalue–torsion products

Henry Zweiman

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## Abstract

We optimize the product of the first Dirichlet eigenvalue and a power of torsional rigidity on a fixed Euclidean ball, allowing the anisotropic energy to vary over all normalized symmetric seminorms. For every dimension  $n \geq 2$  and every energy exponent  $1 < p < \infty$ , we prove that the Euclidean norm uniquely maximizes the product for all sufficiently small positive torsion exponents and uniquely minimizes it for all sufficiently large exponents. In either regime, the product deficit is equivalent to the spherical average of  $1 - H^p$ . Consequently it controls the supremum norm distance from the Euclidean norm with the optimal power  $(n+1)/2$ . The proof combines radial variational tests, convex symmetrization, and an integral comparison between support and radial deficits of convex bodies inside a ball. For quadratic growth, this answers the ball cases of two Euclidean-optimality questions of Buttazzo and Fernandes Horta, beyond the quadratic-seminorm class. No regularity or strict convexity of the competing anisotropies is assumed.

**Keywords:** anisotropic spectral optimization; torsional rigidity; nonlinear eigenvalue; convex symmetrization; quantitative stability.

## 1. Introduction and main results

Fix the Euclidean unit ball  $B \subset \mathbb{R}^n$ , where  $n \geq 2$ , and an exponent  $1 < p < \infty$ . Denote its volume by  $V$ , write  $S = \partial B$ , and let  $\sigma$  be probability surface measure on  $S$ . We consider the class  $\mathcal{H}$  of symmetric seminorms  $H : \mathbb{R}^n \rightarrow [0, \infty)$  normalized by

$$\max_{\theta \in S} H(\theta) = 1. \quad (1.1)$$

Thus  $H$  is convex, satisfies  $H(t\xi) = |t|H(\xi)$ , and may vanish on a nonzero subspace. The Euclidean norm is denoted by  $E(\xi) = |\xi|$ . For  $H \in \mathcal{H}$  define

$$\begin{aligned} \lambda_H &= \inf_{u \in W_0^{1,p}(B) \setminus \{0\}} \frac{\int_B H(\nabla u)^p dx}{\int_B |u|^p dx}, \\ T_H &= \left( \sup_{u \in W_0^{1,p}(B) \setminus \{0\}} \frac{|\int_B u dx|^p}{\int_B H(\nabla u)^p dx} \right)^{1/(p-1)}. \end{aligned} \quad (1.2)$$

The dependence on  $n, p$  is suppressed. These quantities are finite and strictly positive even for degenerate seminorms; a direct proof is given below. The definition does not require attainment in the

degenerate case. Our torsion convention agrees with the integral of the solution of the torsion equation in the usual coercive setting. In particular,  $T_H^{p-1}$  is the variational quotient before taking the root in (1.2).

The normalized product and its angular deficit are

$$R_q(H) = \frac{\lambda_H T_H^q}{\lambda_E T_E^q}, \quad \delta_p(H) = \int_S (1 - H(\theta)^p) d\sigma(\theta). \quad (1.3)$$

The exponent  $q > 0$  is independent of the energy exponent  $p$ . Since  $H \leq E$ , decreasing the anisotropy lowers the eigenvalue and raises the torsion. Separate monotonicity statements therefore do not determine the product. Even convergence of optimizers to  $E$  as  $q$  tends to an endpoint would not establish that  $E$  is the exact optimizer at a fixed finite  $q$ .

**Theorem 1.1 (Euclidean persistence and deficit equivalence).** For every  $n \geq 2$  and  $1 < p < \infty$ , there are numbers

$$0 < q_-(n, p) < p - 1 < q_+(n, p) < \infty \quad (1.4)$$

with the following properties. The Euclidean norm uniquely maximizes  $\lambda_H T_H^q$  over  $\mathcal{H}$  whenever  $0 < q \leq q_-$ , and uniquely minimizes it whenever  $q \geq q_+$ . For each fixed  $q$  in these regimes, set

$$D_q(H) = \begin{cases} 1 - R_q(H), & 0 < q \leq q_-, \\ R_q(H) - 1, & q \geq q_+. \end{cases} \quad (1.5)$$

There are constants  $0 < a_{n,p,q} \leq b_{n,p,q} < \infty$  such that

$$a_{n,p,q} \delta_p(H) \leq D_q(H) \leq b_{n,p,q} \delta_p(H) \quad (H \in \mathcal{H}). \quad (1.6)$$

The theorem applies to the full infinite-dimensional class of normalized symmetric seminorms, with no ellipticity constant prescribed in advance. In particular it includes polyhedral norms and seminorms of every positive rank. Explicit, though nonoptimal, choices of the thresholds are provided below.

The associated uniform distance is

$$\varepsilon(H) = \|H - E\|_{L^\infty(S)} = \max_{\theta \in S} (1 - H(\theta)). \quad (1.7)$$

**Theorem 1.2 (optimal stability power).** Under the assumptions and exponent restrictions of Theorem 1.1, there is  $c_{n,p,q} > 0$  such that

$$D_q(H) \geq c_{n,p,q} \varepsilon(H)^{(n+1)/2} \quad (H \in \mathcal{H}). \quad (1.8)$$

The power  $(n+1)/2$  cannot be replaced by a smaller one, even if the competitors are restricted to genuine norms. More precisely, there are norms  $H_\varepsilon \in \mathcal{H}$ , for  $0 < \varepsilon < 1/2$ , with  $\varepsilon(H_\varepsilon) = \varepsilon$  and

$$D_q(H_\varepsilon) \leq C_{n,p,q} \varepsilon^{(n+1)/2}. \quad (1.9)$$

By scaling the spatial variable, the same normalized conclusions hold on any Euclidean ball. The thresholds do not depend on its radius: the eigenvalue scales by the inverse  $p$ -th power of the radius and torsion by the power  $n + p/(p-1)$ , and these factors cancel in  $R_q$ .

## Relation to previous work

Buttazzo and Fernandes Horta [BFH] study eigenvalue-torsion products while varying normalized seminorms on a fixed domain. Their Section 6 asks whether the norm optimizers in the small- $q$  maximum and large- $q$  minimum regimes are Euclidean. For balls, their Theorem 4.10 proves maximality for  $q \leq 1$  within the class of quadratic seminorms. Theorem 1.1 answers both questions on balls in the full symmetric-seminorm class, and treats the general  $p$ -growth setting also proposed in their final remarks. It does not resolve their questions for arbitrary fixed domains or determine the optimal thresholds.

The optimization of the eigenvalue alone, including rigidity of Euclidean maximality, was studied by Fernandes Horta and Montenegro [HM]; see their Theorem 1.5. We do not claim that endpoint result as new. Our spectral lower estimate uses the convex symmetrization theorem of Alvino, Ferone, Trombetti and Lions [AFTL]. The body and normalization conventions for this input are stated explicitly in Section 4. The convex-geometric part uses support and radial deficits, quantities related to the intrinsic and dual volume deviations studied in [BHK]. We provide the geometric estimate needed here in full and make no separate priority claim for the elementary cap geometry.

The main analytic point is a common linear control, in  $\delta_p(H)$ , of the spectral loss and torsion gain near  $E$ . Directional first variation alone gives no such uniform estimate over increasingly concentrated angular perturbations. The support-to-radial comparison in Section 3 supplies this missing control without smoothness assumptions. It also produces the sharp supremum norm stability power.

## 2. Convex representation and global variational bounds

Every  $H \in \mathcal{H}$  is the support function of the compact centrally symmetric convex set

$$K = \{x \in \mathbb{R}^n : x \cdot \xi \leq H(\xi) \text{ for every } \xi \in \mathbb{R}^n\}, \quad H = h_K. \quad (2.1)$$

Indeed, the support representation follows from separation for finite continuous sublinear functions. Normalization gives  $K \subset \overline{B}$  and  $K \cap S \neq \emptyset$ . To check the latter, the maximum of  $h_K$  on  $S$  equals the maximum of  $|x|$  over  $K$ . In particular, there is  $v \in K \cap S$ , and symmetry implies

$$[-v, v] \subset K, \quad H(\xi) \geq |v \cdot \xi|. \quad (2.2)$$

Also  $|H(\xi) - H(\eta)| \leq |\xi - \eta|$ . When  $K$  has interior, write  $\rho_K$  for its radial function and  $H^\circ$  for the dual norm of  $H$ . Then

$$K = \{H^\circ \leq 1\}, \quad H^\circ(\theta) = \rho_K(\theta)^{-1}. \quad (2.3)$$

Thus  $K$  is the polar of the unit ball of  $H$ , not the unit ball of  $H$  itself.

**Lemma 2.1 (global bounds).** Put

$$m_H = \int_S H^p d\sigma = 1 - \delta_p(H), \quad p' = \frac{p}{p-1}. \quad (2.4)$$

For every  $H \in \mathcal{H}$ ,

$$2^{-p} \leq \lambda_H \leq m_H \lambda_E, \quad m_H^{-1/(p-1)} T_E \leq T_H \leq V 2^{p'}. \quad (2.5)$$

Moreover,  $m_H \geq \int_S |\theta \cdot v|^p d\sigma > 0$ , where this last integral depends only on  $n, p$ .

*Proof.* For  $f \in C_c^\infty(I)$  on an interval of length  $L \leq 2$ , the fundamental theorem of calculus and Hölder's inequality imply

$$\int_I |f|^p \leq L^p \int_I |f'|^p \leq 2^p \int_I |f'|^p. \quad (2.6)$$

Apply this on chords parallel to  $v$  and integrate over the orthogonal variables. Using (2.2) and density in  $W_0^{1,p}(B)$  gives

$$\int_B |u|^p \leq 2^p \int_B H(\nabla u)^p. \quad (2.7)$$

This proves the eigenvalue lower bound. Combining (2.7) with  $|\int_B u|^p \leq V^{p-1} \int_B |u|^p$  proves the torsion upper bound in (2.5). In particular, none of the denominators in (1.2) vanish for nonzero admissible functions.

A Euclidean first eigenfunction can be chosen radial and nonnegative by Schwarz symmetrization of a minimizer. Existence follows by the direct method and the compact embedding into  $L^p(B)$ . For every radial Sobolev function  $f$ , homogeneity and polar coordinates give

$$\int_B H(\nabla f)^p = m_H \int_B |\nabla f|^p. \quad (2.8)$$

Testing the eigenvalue with a radial Euclidean minimizer proves its upper bound. The Euclidean torsion function and its integral are

$$w_E(x) = \frac{n^{-1/(p-1)}}{p'} (1 - |x|^{p'}), \quad T_E = \frac{V n^{-1/(p-1)}}{n + p'}. \quad (2.9)$$

In fact  $|\nabla w_E|^{p-2} \nabla w_E = -x/n$ , so  $-\Delta_p w_E = 1$  in distributions. Testing this identity with  $w_E$  gives

$$\int_B |\nabla w_E|^p = \int_B w_E = T_E. \quad (2.10)$$

Hölder applied to the same weak identity shows that  $w_E$  attains the Euclidean torsion quotient. Equations (2.8)-(2.10) then give its lower bound in (2.5). The final assertion follows by integrating (2.2) on  $S$ .  $\square$

We shall use the fixed constants

$$\ell = \min \left\{ \frac{1}{2}, \frac{2^{-p}}{\lambda_E} \right\}, \quad M = \max \left\{ 2, \frac{V 2^{p'}}{T_E} \right\}. \quad (2.11)$$

They give the global bounds  $\lambda_H/\lambda_E \geq \ell$  and  $T_H/T_E \leq M$ . Their sharp values are unnecessary.

### 3. An integral comparison of support and radial deficits

For  $\nu \in S$  and  $r > 0$ , let  $C(\nu, r) = \{\theta \in S : |\theta - \nu| < r\}$  be a chordal cap. The notation  $aC$  denotes the cap with the same center and radius multiplied by  $a$ .

**Lemma 3.1 (support-to-radial comparison).** There is  $A_n < \infty$  such that every compact convex set  $K$  with  $\frac{1}{2}\bar{B} \subset K \subset \bar{B}$  satisfies

$$\int_S (1 - \rho_K) d\sigma \leq A_n \int_S (1 - h_K) d\sigma. \quad (3.1)$$

Symmetry is not required in this lemma.

*Proof.* We first record a cap estimate. Suppose that  $h_K(\nu) = 1 - e$ , where  $0 < e \leq 1$ . Since  $K \subset \{x \in \bar{B} : x \cdot \nu \leq 1 - e\}$ , its support function at a direction  $\eta$  making angle  $\alpha \leq \sqrt{e}/4$  with  $\nu$  satisfies

$$\begin{aligned} h_K(\eta) &\leq (1 - e) \cos \alpha + \sqrt{2e - e^2} \sin \alpha \\ &\leq 1 - e + \frac{\sqrt{2}}{4} e \leq 1 - \frac{e}{2}. \end{aligned} \quad (3.2)$$

For completeness,  $\cos \alpha > 1 - e$  because  $1 - \cos \alpha \leq \alpha^2/2 \leq e/32$ . Thus the unconstrained support point  $\eta$  is excluded by the cutting plane, and maximizing the linear functional on the truncated ball puts the maximizing point on that plane. Decomposition into the  $\nu$  and  $\nu^\perp$  components gives the first line of (3.2). The last estimates use  $1 - e \geq 0$  and  $\sin \alpha \leq \alpha$ .

Set  $r = \sqrt{e}/8$ . On  $C(\nu, r)$  the angle is less than  $2r \leq \sqrt{e}/4$ , so this cap lies in  $\{1 - h_K \geq e/2\}$ .

Now fix  $\theta \in S \setminus K$ . Let  $z$  be its metric projection onto  $K$ , and put

$$d = |\theta - z|, \quad \nu = \frac{\theta - z}{d}, \quad e = 1 - h_K(\nu). \quad (3.3)$$

The projection inequality gives  $z \cdot \nu = h_K(\nu)$ , hence  $\theta \cdot \nu = 1 - e + d$ . Therefore

$$0 < d \leq e \leq \frac{1}{2}, \quad |\theta - \nu|^2 = 2(e - d) \leq 2e. \quad (3.4)$$

The cap  $C = C(\nu, \sqrt{e}/8)$  from (3.2) consequently has  $\theta \in 12C$ .

Fix  $t > 0$  and write  $E_t = \{\theta \in S : \text{dist}(\theta, K) > t\}$ . For each  $\theta \in E_t$  choose the cap just constructed; its associated  $e$  exceeds  $t$ . Every compact subset  $L \subset E_t$  is covered by finitely many of the open caps  $12C$ . Order the corresponding original caps by decreasing radius, and retain a cap precisely when it is disjoint from every previously retained cap. The retained caps  $C_i$  are disjoint. Any discarded cap of radius  $r$  intersects a retained cap of radius  $r_i \geq r$ , so its twelvefold enlargement lies in  $14C_i$ , and hence in  $30C_i$ . Thus  $L \subset \bigcup_i 30C_i$ .

Spherical cap measure is doubling: there is  $D_n < \infty$  such that

$$\sigma(C(\nu, 30r)) \leq D_n \sigma(C(\nu, r)) \quad \left(0 < r \leq \frac{1}{8\sqrt{2}}\right). \quad (3.5)$$

This follows directly by writing cap area in polar coordinates. The ratio tends to  $30^{n-1}$  as  $r \downarrow 0$  and is bounded on every remaining compact interval; a cap whose chordal radius exceeds 2 is the entire sphere. Each retained cap is contained in  $\{1 - h_K > t/2\}$  by (3.2). Disjointness and (3.5) give

$$\sigma(L) \leq D_n \sum_i \sigma(C_i) \leq D_n \sigma\{1 - h_K > t/2\}. \quad (3.6)$$

By inner regularity this holds with  $L = E_t$ . Integrating in  $t$  yields

$$\int_S \text{dist}(\theta, K) d\sigma \leq 2D_n \int_S (1 - h_K) d\sigma. \quad (3.7)$$

The gauge  $g_K$  is 2-Lipschitz since  $K$  contains  $\frac{1}{2}\overline{B}$ . Indeed, subadditivity gives  $g_K(x) - g_K(y) \leq g_K(x - y) \leq 2|x - y|$ , and one can interchange  $x, y$ . For the projection  $z$  above,  $g_K(z) \leq 1$ , so

$$\rho_K(\theta)^{-1} - 1 \leq g_K(\theta) - g_K(z) \leq 2 \text{dist}(\theta, K). \quad (3.8)$$

Since  $0 < \rho_K \leq 1$ , we have  $1 - \rho_K \leq \rho_K^{-1} - 1$ . Equations (3.7)-(3.8) prove (3.1) with  $A_n = 4D_n$ .  $\square$

The estimate transfers angular support loss to the radial moments needed by both variational problems. A supremum norm comparison alone would not give the linear dependence on  $\delta_p$  used below.

## 4. Uniform estimates near the Euclidean norm

Set

$$\eta_0 = \frac{1}{2}(1 - (3/4)^p)\sigma(C(\nu, 1/4)), \quad (4.1)$$

which is positive and independent of  $\nu$ . If  $\min_S H \leq 1/2$ , the 1-Lipschitz property implies  $H \leq 3/4$  on a cap of radius  $1/4$ . Thus  $\delta_p(H) \geq 2\eta_0$ . In particular, if  $\delta_p(H) \leq \eta_0$ , then  $H > 1/2$  on  $S$  and  $K$  contains  $\frac{1}{2}\overline{B}$ .

For the remainder of this section write  $\delta = \delta_p(H) \leq \eta_0$ . Since  $1 - H \leq 1 - H^p$  on  $S$ , Lemma 3.1 gives

$$\int_S (1 - \rho_K) d\sigma \leq A_n \delta. \quad (4.2)$$

Polar integration and  $1 - r^n \leq n(1 - r)$  for  $0 \leq r \leq 1$  yield

$$\frac{|K|}{V} = \int_S \rho_K^n d\sigma \geq 1 - nA_n \delta. \quad (4.3)$$

Similarly, on  $[1/2, 1]$  the derivative of  $r^{-p'}$  has magnitude at most  $p'2^{p'+1}$ , and therefore

$$\int_S (H^\circ)^{p'} d\sigma \leq 1 + p'2^{p'+1}A_n \delta. \quad (4.4)$$

Fix

$$C = \max\{nA_n, p'2^{p'+1}A_n, 1\}, \quad \eta = \min\{\eta_0, (2C)^{-1}, 1/4\}. \quad (4.5)$$

**Lemma 4.1 (spectral and torsion control).** If  $\delta_p(H) \leq \eta$ , then

$$(1 - C\delta_p(H))^{p/n} \leq \frac{\lambda_H}{\lambda_E} \leq 1 - \delta_p(H), \quad (4.6)$$

and

$$(1 - \delta_p(H))^{-1/(p-1)} \leq \frac{T_H}{T_E} \leq 1 + C\delta_p(H). \quad (4.7)$$

*Proof.* Only the lower spectral and upper torsion bounds remain. Convex symmetrization [AFTL, Theorem 3.1 and equation (3.5)] implies, for  $u \in W_0^{1,p}(B)$  and its Euclidean Schwarz rearrangement  $u^\#$ ,

$$\int_B H(\nabla u)^p dx \geq \left(\frac{|K|}{V}\right)^{p/n} \int_B |\nabla u^\#|^p dx. \quad (4.8)$$

Here one first replaces  $u$  by  $|u|$ ; evenness of  $H$  preserves the energy. To specify the normalization in (4.8), let  $\mu(t) = |\{|u| > t\}|$ . The energy of the convex rearrangement has the level-set expression

$$\int_0^\infty [n|K|^{1/n}\mu(t)^{1-1/n}]^p (-\mu'(t))^{1-p} dt, \quad (4.9)$$

in the usual rearrangement interpretation. The corresponding expression for  $u^\#$  replaces  $|K|$  by  $V$ . The convex rearrangement theorem decreases the original energy; comparison of these expressions gives (4.8), with the general Sobolev statement supplied by that theorem. The body denoted  $K$  in Section 2 of [AFTL] is the unit ball of  $H$ ; its polar is the body called  $K$  here. Their auxiliary volume normalization can be removed by scaling  $H$ : both sides of (4.8) scale by its  $p$ -th power. Their hypotheses allow nonsmooth convex norms with positive coercivity, which holds in the present neighborhood. No regularity approximation of  $H$  or equality case is needed.

Equimeasurability preserves the  $L^p$  denominator in the Rayleigh quotient. Taking the infimum in (4.8) gives  $\lambda_H/\lambda_E \geq (|K|/V)^{p/n}$ , and (4.3) proves (4.6).

For torsion use the Euclidean flux

$$F = |\nabla w_E|^{p-2} \nabla w_E = -\frac{x}{n}. \quad (4.10)$$

For every  $u \in W_0^{1,p}(B)$ , integration by parts, duality of norms and Hölder's inequality give

$$\left| \int_B u \right|^p = \left| \int_B F \cdot \nabla u \right|^p \leq \left( \int_B H^\circ(F)^{p'} \right)^{p-1} \int_B H(\nabla u)^p. \quad (4.11)$$

Take the supremum and the power  $1/(p-1)$  in (1.2). Radiality of  $F$  and (2.10) imply

$$T_H \leq \int_B H^\circ(F)^{p'} = T_E \int_S (H^\circ)^{p'} d\sigma. \quad (4.12)$$

Equation (4.4) proves the upper torsion bound. Notice that the duality test uses the flux  $F$ , whose Euclidean  $p'$ -energy equals  $T_E$ , rather than the gradient  $\nabla w_E$  with an incorrect exponent.  $\square$

**Corollary 4.2.** For  $q > 0$  and  $\delta = \delta_p(H) \leq \eta$ ,

$$\log R_q(H) \leq (-1 + qC)\delta, \quad \log R_q(H) \geq \left( \frac{q}{p-1} - \frac{2pC}{n} \right) \delta. \quad (4.13)$$

*Proof.* For the first estimate use  $\log(1 - \delta) \leq -\delta$  and  $\log(1 + C\delta) \leq C\delta$ . For the second use  $\log(1 - C\delta) \geq -2C\delta$ , valid because  $C\delta \leq 1/2$ , and  $-\log(1 - \delta) \geq \delta$ . Apply (4.6)-(4.7) to the logarithm of the product.  $\square$

## 5. Persistence and quantitative equivalence

With the constants from (2.11) and (4.5), choose

$$q_- = \min \left\{ \frac{p-1}{2}, \frac{1}{2C}, \frac{-\log(1-\eta)}{2\log M} \right\}, \quad (5.1)$$

and

$$q_+ = (p-1) \max \left\{ \frac{2pC}{n} + 1, \frac{\log(1/\ell) + 1}{-\log(1-\eta)} \right\}. \quad (5.2)$$

These constants satisfy (1.4).

*Proof of Theorem 1.1.* Suppose first that  $0 < \delta_p(H) \leq \eta$ . Corollary 4.2 gives

$$R_q(H) \leq e^{-\delta_p(H)/2} < 1 \quad (0 < q \leq q_-), \quad (5.3)$$

and

$$R_q(H) \geq e^{\delta_p(H)} > 1 \quad (q \geq q_+). \quad (5.4)$$

For  $\delta_p(H) \geq \eta$ , Lemma 2.1 and (2.11) imply

$$R_q(H) \leq (1-\eta)M^q \leq \sqrt{1-\eta} < 1 \quad (0 < q \leq q_-), \quad (5.5)$$

whereas

$$R_q(H) \geq \ell(1-\eta)^{-q/(p-1)} \geq e > 1 \quad (q \geq q_+). \quad (5.6)$$

These estimates include all degenerate seminorms. Finally,  $\delta_p(H) = 0$  forces  $H = 1$  everywhere on  $S$  by continuity, so  $H = E$ . Equations (5.3)-(5.6) prove uniqueness and existence of the stated optimizers directly.

For completeness, we establish both sides of (1.6). In the small- $q$  regime and for  $\delta = \delta_p(H) \leq \eta$ , (5.3) gives

$$D_q(H) \geq 1 - e^{-\delta/2} \geq \frac{1}{2}e^{-1/2}\delta. \quad (5.7)$$

Since  $T_H \geq T_E$ , the other side follows from (4.6):

$$R_q(H) \geq (1 - C\delta)^{p/n} \geq 1 - \max\{1, p/n\}C\delta. \quad (5.8)$$

The last inequality follows from Bernoulli's inequality if  $p/n \geq 1$  and from  $(1-x)^{p/n} \geq 1-x$  otherwise. In the large- $q$  regime, (5.4) implies  $D_q(H) \geq \delta$ . For its upper bound,  $\lambda_H \leq \lambda_E$  and (4.7) give

$$R_q(H) \leq (1 + C\delta)^q \leq 1 + C_q\delta, \quad (5.9)$$

where one may take  $C_q = qC \max\{1, (3/2)^{q-1}\}$ .

It remains to treat  $\delta \geq \eta$ . We always have  $\delta < 1$ . The uniform gaps in (5.5)-(5.6) therefore give a positive constant times  $\delta$  as a lower bound for  $D_q$ . In the maximum regime  $D_q \leq 1 \leq \delta/\eta$ . In the minimum regime  $R_q \leq M^q$ , so  $D_q \leq M^q\delta/\eta$ . Combining these constants with (5.7)-(5.9) proves (1.6).  $\square$

## 6. Optimal supremum norm stability

**Lemma 6.1 (angular concentration bound).** There is  $c_n > 0$  such that every  $H \in \mathcal{H}$  satisfies

$$c_n \varepsilon(H)^{(n+1)/2} \leq \delta_p(H) \leq p\varepsilon(H). \quad (6.1)$$

*Proof.* If  $\varepsilon = 0$ , both assertions are immediate. Otherwise choose  $\nu \in S$  with  $H(\nu) = 1 - \varepsilon$ . The cap calculation (3.2), which holds for every  $0 < e \leq 1$  without an inball assumption, applies with  $e = \varepsilon$ . On  $C(\nu, \sqrt{\varepsilon}/8)$  we have  $1 - H \geq \varepsilon/2$ , and hence  $1 - H^p \geq \varepsilon/2$ . Its measure is at least a dimensional constant times  $\varepsilon^{(n-1)/2}$ . This proves the lower bound. The upper bound follows from  $1 - t^p \leq p(1 - t)$  for  $0 \leq t \leq 1$ .  $\square$

*Proof of Theorem 1.2.* The lower bound follows at once by combining (1.6) and Lemma 6.1. For sharpness, take

$$K_\varepsilon = \overline{B} \cap \{x : |x \cdot e_1| \leq 1 - \varepsilon\}, \quad H_\varepsilon = h_{K_\varepsilon}, \quad 0 < \varepsilon < \frac{1}{2}. \quad (6.2)$$

This centrally symmetric body contains  $(1 - \varepsilon)\overline{B}$  and, since  $n \geq 2$ , touches  $S$  on the equator perpendicular to  $e_1$ . Consequently  $H_\varepsilon$  is a genuine normalized norm. At  $e_1$  its support equals  $1 - \varepsilon$ , so  $\varepsilon(H_\varepsilon) = \varepsilon$ .

Whenever  $|\theta \cdot e_1| \leq 1 - \varepsilon$ , the point  $\theta$  itself belongs to  $K_\varepsilon$ , and thus  $H_\varepsilon(\theta) = 1$ . The angular deficit is supported on two spherical caps of angular radius  $\arccos(1 - \varepsilon) \leq C\sqrt{\varepsilon}$ . Their total measure is at most  $C_n \varepsilon^{(n-1)/2}$ . Also  $1 - H_\varepsilon^p \leq p\varepsilon$ , whence

$$\delta_p(H_\varepsilon) \leq C_{n,p} \varepsilon^{(n+1)/2}. \quad (6.3)$$

The upper bound in (1.6) proves (1.9). If a uniform estimate  $D_q(H) \geq c\varepsilon(H)^\gamma$  held for some  $\gamma < (n+1)/2$ , evaluating it at  $H_\varepsilon$  and letting  $\varepsilon \downarrow 0$  would force a fixed positive  $c$  to be zero. This proves optimality.  $\square$

## 7. Scope and further questions

The argument identifies exact optimizers at finite exponents and gives a common angular measure of their stability. The two regimes and the nonlinear extension follow from the same comparison mechanism. None of the constants in the construction is intended to be optimal.

The natural next question is to determine the full interval of  $q$  for which  $E$  is extremal. For  $p = 2$ , the quadratic-seminorm conclusion of [BFH, Theorem 4.10] reaches  $q = 1$  in the maximizing problem; extending that endpoint to arbitrary norms requires information not supplied by (4.13). Another question is whether analogous persistence holds on fixed domains other than balls. Our radial trial identities and dual flux calculation depend on the ball geometry, and the present theorem makes no assertion for general domains. Finally, constants uniform as  $p \downarrow 1$  or  $p \uparrow \infty$  are not obtained here.

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